

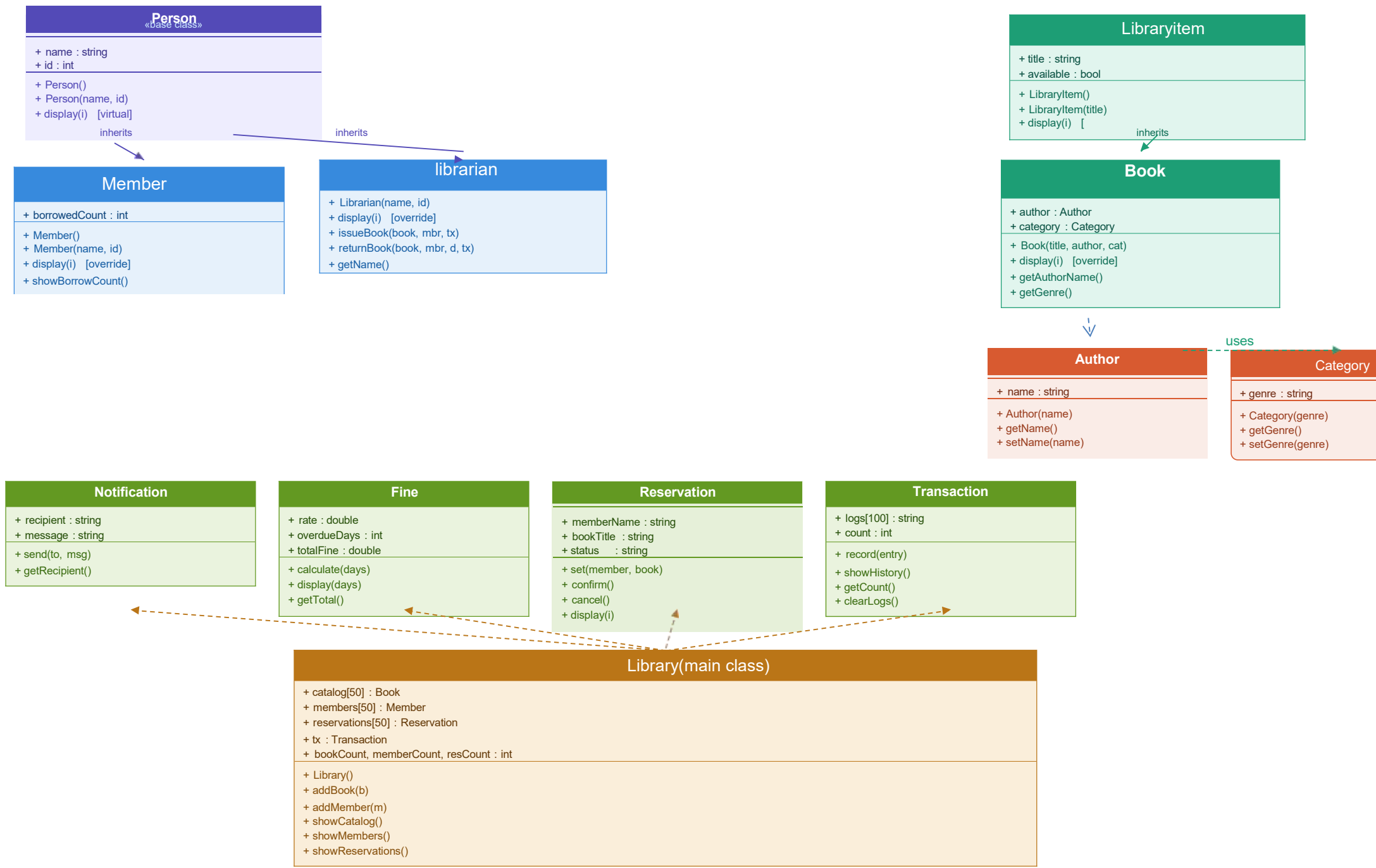
Object oriented programming



Submitted by: khubaib gul(094) & Zaid saleh(085)

Submitted to:DR.Saba Mehmood

Library Management System — OOP Class Diagram



Legend

—> Solid arrow — Inheritance

- - -> Dashed arrow — Composition / uses

- - - -> Dashed arrow — Aggregation / manages

Classes Used:

- **Person** (*Person.h*) : Base class for people; has name and id
- **Member** (*Member.h*) : Derived from Person; tracks borrowedCount
- **Librarian** (*Librarian.h*) : Derived from Person; handles issue/return logic
- **LibraryItem** (*LibraryItem.h*) : Base class for items; has title and available
- **Book** (*Book.h*) : Derived from LibraryItem; has an Author and Category
- **Author** (*Author.h*) : Simple class holding an author's name
- **Category** (*Category.h*) : Simple class holding a genre
- **Library** (*Library.h*) : Main container ; holds books, members, reservations, and a transaction log
- **Reservation** (*Reservation.h*) : Tracks a book reservation with memberName, bookTitle, and status
- **Transaction** (*Transaction.h*) : Logs issue/return history as strings
- **Notification** (*Notification.h*) : Sends a simple console notification to a member